

Course content and style

AI-Assisted Data Analysis in Marketing

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

Dr Arne Floh, Senior Lecturer in Marketing
WU Vienna, Institute for International Business

arne.floh@wu.ac.at
Office hours: book online (Google Calendar)

WINTER TERM 2026/27

The journey begins



Module Convenor – Dr Arne Floh



Personal and contact information

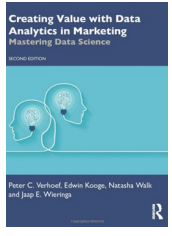
- **Senior Lecturer in International Marketing – WU Vienna**
- Deputy Program Director MSc ExInt
  Export- und Internationalisierungsmanagement
- WWW: wu.ac.at/en/welthandel/arne-floh
- LinkedIn: http://www.linkedin.com/arnefloh

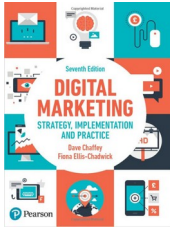
- T: 01 31336 6367
- E: arne.floh@wu.ac.at
- Room D1.5.068
- Office hours:
  book online (Google Calendar)

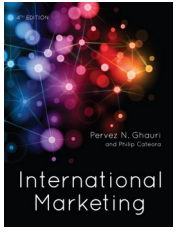
Module Convenor – Dr Arne Floh

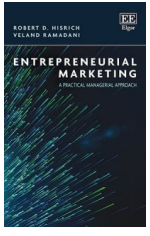


- Work Experience
  - International & Industry Background
  - Teaching Experience > 25 Years
- Teaching
  - International Marketing
  - International Business Projects
  - Marketing Research and Marketing Analytics
  - Global Branding
  - Supervision (BSc, MSc, PhD)
- Research
  - Dynamic Capabilties and Firm Performance
  - GenAI and Marketing
  - SMEs and Internationalization
- Private
  - Football
  - Electronic Gadgets
  - Austrian Wine

























Module content

Data-driven marketing decisions: measuring, managing and analysing marketing performance data to plan markets, improve effectiveness and optimise return on investment (ROI).

International focus: from market entry and customer retention to the marketing mix and risk, analytics gives the evidence for decisions across countries.

Python throughout: data handling, statistics, charts and machine learning. No prior Python knowledge needed; we start from zero.

AI-assisted: GitHub Copilot for Education writes much of the code with you; you check every result and own every number you report.

Six questions this course answers

1	**How effective is my advertising and promotional spending?** cross-sectional regression, simple marketing mix model	Session 2
2	**How should I set prices and understand price sensitivity?** regression, price experiments	Sessions 2, 3
3	**How do I design campaigns and measure their effect?** experiments, A/B tests	Session 3
4	**How should I allocate my budget over time?** across channels and countries: advanced marketing mix modelling	Session 4
5	**Are all my customers profitable?** CLV, RFM, segmentation	Session 5
6	**How can I increase customer engagement?** segments, churn with logistic regression	Session 5

Course roadmap: six sessions

1	14 Oct	**Kick-off**	online on Zoom: course, set-up, GitHub Copilot
2	21 Oct	**Sales forecast I**	EDA, simple marketing mix model with dummy variables
3	28 Oct	**Experiments**	A/B tests, hypothesis tests
4	4 Nov	**Sales forecast II**	advanced marketing mix model over time
5	11 Nov	**Customer analytics**	CLV, RFM, segmentation, churn
6	18 Nov	**Course work**	group case, guest speaker

Along the way: online certificates, four coding exercises, online coaching

18 Nov, 16:00: written exam

Learning outcomes

After this module, you will be able to:

Understand

1. **Explain marketing analytics:** concepts, tools, international context
2. **Know the data:** customer, transaction and social media data

Apply

3. **Explore data (EDA):** summarise and visualise
4. **Segment customers** with RFM and clustering
5. **Predict** churn and sales with logistic regression and ML
6. **Optimise the marketing mix** with regression and attribution
7. **Test hypotheses,** including A/B tests

Transfer

8. **Communicate insights** to technical and non-technical audiences
9. **Work in teams** on data-based decisions
10. **Think critically** about data, methods and results

Methods of teaching and learning

Every lab session (Sessions 2 to 5)

Concept input → **Guided lab with Copilot** → **Team exercise** → **Quiz and coding task**

Between sessions

- Preparatory reading
- Online lectures and expert talks
- Online certificates and coding exercises
- Online coaching sessions

In the classroom

- Active participation
- Individual and group work
- Case studies and a guest speaker
- Laptop with Python, Positron, Copilot

Attendance: blended learning, so you may miss at most one in-person session.

Module roadmap

Date	Topic	Methods
14 Oct 08:30–10:00	**Kick-off**	Online on Zoom. Course content and style; set-up: Python, Positron, Quarto, GitHub Desktop, GitHub Copilot for Education
21 Oct 08:30–12:30	**Sales forecast I**	Exploratory data analysis; cross-sectional sales forecast: simple marketing mix model, regression with dummy variables
28 Oct 08:30–12:30	**Experiments**	Experimental design, A/B testing, hypothesis tests
4 Nov 08:30–12:30	**Sales forecast II**	Longitudinal sales forecast: advanced marketing mix model (trend, seasonality, carry-over, saturation)
11 Nov 08:30–12:30	**Customer analytics**	Customer lifetime value (CLV), RFM, customer segmentation, churn with logistic regression
18 Nov 08:30–12:30	**Course work**	Group case study and guest speaker
18 Nov 16:00–17:30	**Written exam**	Room D1.1.078

Assessment strategy



10 %

Online certificates
2 × 5 %
Individual

20 %

Coding exercises
4 × 5 %
Individual

30 %

Written exam
18 Nov, 16:00
Individual

30 %

Case study report
written report
Group

10 %

Peer review
within your group
Individual

Individual: certificates, coding exercises, exam and peer review (70 %).
Group: the case study report (30 %).
AI tools: expected, with a prompt log; you own every number you report.

Software and set-up

Python 3.14
the language for all analyses

Positron
write and run code, see results

Quarto
code and text into reports

GitHub Desktop
clone and pull the course files

GitHub Copilot
AI coding assistant, free for students

Set-up in Session 1: five steps

1. **Register accounts**
2. **Install the tools**
3. **Clone the repo**
4. **Connect Copilot**
5. **Run the check**

Session 1 on Zoom: join from your laptop, not your phone.
Before that: apply for GitHub Education with your WU e-mail address.

AI in this course: GitHub Copilot

What it does

- Suggests code as you type
- Explains code and errors
- Drafts charts and models

How we use it

- Prompt: say what you need
- Check: run, read, compare
- Explain: in your own words

Your part

- Know the method first
- Keep a short prompt log
- No personal data

Get it free

Create a GitHub account → **Apply: GitHub Education** → **Verify as a WU student** → **Sign in from Positron**

The one rule: AI writes the code, you own the result. Every number in your work must come from code you have run and can explain.

Online certificates

2 — **Online courses.** Short self-paced courses that build the basics for the labs, each ending with a certificate.

5 % — **Per certificate.** Upload each certificate on Canvas by its deadline.

10 % — **In total.** Together the two certificates count one tenth of the final grade.

Platform and deadlines: announced on Canvas. Start early; the courses build the Python basics you need for the labs.

Coding exercises

4
exercises, Sessions 2 to 5

5 %
per exercise

20 %
of the final grade

How it works

Guided lab in class → **Apply the code to a new task** → **Submit on Canvas** → **Feedback and grade**

Rules

- Submit your Quarto file; it must run from top to bottom
- GitHub Copilot is allowed; add a short prompt log at the end
- Individual work: discuss ideas, but write and check your own code

Written exam

30 %
of the final grade

18 Nov
2026, 16:00 to 17:30

D1.1.078
exam room

Format

- Individual, closed book
- In the exam room
- Details on Canvas

Content

- Sessions 1 to 5
- Concepts
- Interpreting model output

Preparation

- Quizzes (not graded)
- Session materials
- Online coaching

Group case study: Alpvia Outdoor

max. 4 students per group	**30 %** case study report	**10 %** peer review	**30 Nov** deadline, 23:59

Four questions from Alpvia's CMO

A	How effective is our advertising and promotion in the stores?	Session 2
B	Did the lower free-shipping threshold pay off?	Session 3
C	How should we spend media over time, and what are Q1 2026 sales?	Session 4
D	Which customers matter most, and who is about to leave?	Session 5

Brief, data and template: case-study/ and data/case/ in the course repository

Case study: parts, points and submission

A	**Stores: advertising and promotion** EDA, regression with logs and dummies	Session 2	20 pts
B	**Free-shipping A/B test** tests, confidence intervals, profit, sample size	Session 3	15 pts
C	**Media over time and Q1 forecast** adstock, saturation, forecast test, scenarios	Session 4	25 pts
D	**Customers: value and churn** RFM, k-means, CLV, logistic regression	Session 5	25 pts
E	**Recommendations and report** summary, limitations, reproducibility	all sessions	15 pts

Submit on Canvas: report.qmd and report.html, one upload per group. It must render, and every number comes from code. AI allowed, with a prompt log.

VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

INSTITUTE FOR INTERNATIONAL BUSINESS
Welthandelsplatz 1, 1020 Vienna, Austria

DR ARNE FLOH
Senior Lecturer in Marketing

F +43-31336-6367
arne.floh@wu.ac.at
www.wu.ac.at/en/welthandel/arne-floh
Office hours: book online